

Which Head Does the Work? How Attention-Head Circuit Findings Carry Over Between Related Language Models

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Abstract

Circuit analyses identify the attention heads that carry a computation, and these head-level findings are reused on related models: later checkpoints, fine-tuned descendants, and models trained from the same initialization on other data. We ask in which of these related models the same heads carry the computation. Public model suites that cross initializations with pretraining corpora, a complete public lineage from pretraining to instruction tuning, and controlled pretraining let us vary one relation between models at a time. Head roles follow the initialization. Models trained from the same initialization agree on which head of a layer takes each role, whatever their corpus, and so share a coordinate system for their components. The heads that causally carry a task follow the training run instead. Models that share an initialization but differ slightly in their data carry indirect-object identification on different heads of the same functional classes, whereas a model’s carriers, once settled during pretraining, persist through midtraining, preference tuning and instruction tuning. Controlled pretraining shows how carriers are assigned: candidate heads compete while a role forms, and the data of the run decide between close candidates. Head-level findings thus transfer along a training lineage and role labels within an initialization, while across separately trained models the algorithm transfers and the carriers have to be localized again.

1 Introduction

Mechanistic interpretability explains what a language model computes by locating the components that carry the computation. Its findings are statements about specific heads. The indirect-object identification (IOI) circuit of GPT-2 consists of name-mover heads, heads that suppress the repeated subject, and the heads that feed them (Wang et al., 2023), and in-context copying runs through

a previous-token head composed with an induction head (Olsson et al., 2022). Such statements are increasingly reused beyond the model in which they were found. Circuits are tracked across the checkpoints of a training run (Tigges et al., 2024), compared between a base model and its fine-tuned versions (Prakash et al., 2024), and used as the readout of training-data interventions that retrain a model and examine the same head again (Chen et al., 2026). Each of these uses assumes that a head in one model corresponds to a head in another.

The architecture does not provide this correspondence. Heads within a layer are interchangeable: permuting them together with their output projections leaves the network’s function unchanged, so a head index refers to a component only relative to a training history. Studies of stability across random seeds find that circuits keep their algorithm while the heads that implement it change (Tigges et al., 2024), that individual heads are unstable across seeds (Bali et al., 2026), and that heads with the same role rarely sit at the same position (Trott, 2025). A seed fixes the initialization and the order of the data together, and these studies compare runs that were trained separately, whereas findings are usually transferred to models that share the source model’s initialization or continue its training. Whether a head-level finding holds in such models is the question of this paper.

We answer it by varying one relation at a time. DataDecide trains the same three initializations on 25 pretraining corpora (Magnusson et al., 2025), Pythia trains a standard and a deduplicated model from one initialization at every size (Biderman et al., 2023), and OLMo 2 releases every stage of a lineage from pretraining through midtraining, supervised fine-tuning (SFT), direct preference optimization (DPO) and instruction tuning (Team OLMo et al., 2024). Controlled pretraining, finally, lets us intervene on the data of a single run. In every model we measure nine head roles and the

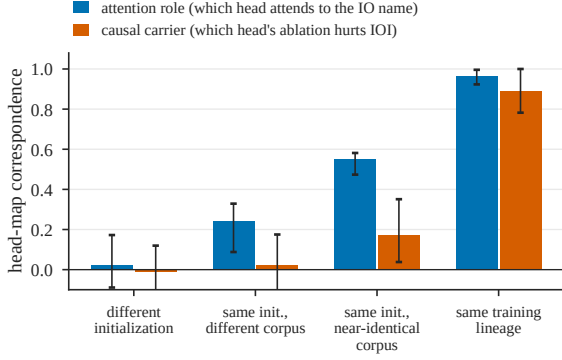


Figure 1: What carries over between two models, by their relation. Blue: correspondence of the attention maps from the final position to the indirect object; or-ange: reliability-corrected correspondence of the causal maps of IOI (single-head ablation). Bars are means over model pairs, whiskers the range. Models from one initialization share the attention role of a head but not its causal role; models in one training lineage share both.

heads that causally carry IOI, and every comparison between models is corrected for the reliability of the measurements.

The answer depends on the relation (Figure 1). Models trained from the same initialization agree on which head of a layer takes each role across all 25 corpora, whereas models from different initializations agree only on the profile over layers (§3). The initialization fixes a coordinate system that all models trained from it share. It does not fix the causal carriers of IOI (§4). Within the shared coordinates, models that differ in their data carry IOI on different heads, even when their corpora differ only by the removal of 1.5% of the data, while running the same algorithm. Along one training run, the carriers move between heads during pretraining and then stay, so the carriers at the end of pretraining are those after midtraining, SFT, DPO and instruction tuning (§5). Controlled pretraining reveals the competition behind this assignment (§6). A role with a single leading head stays on that head when training data are masked during its emergence, while between two close candidates a random mask of a tenth of the data changes the winner.

Our contributions are as follows. (1) We separate shared initialization, different training data and continued training as relations between models, and show that role labels transfer within an initialization, causal carriers only along a training lineage, and the IOI algorithm across separately

Population	Models	Relation varied	Used in
DataDecide-1B	25 corpora $\times$ 3 init.; 69 with verified init.	corpus, init.	§3, §4, §5
DataDecide sizes	4M–750M, 10–23 corpora	corpus, init., scale	§3
Pythia	70M–12B, standard and deduplicated	data (same init.)	§3, §4
Pythia checkpoints	160M, 410M pairs, 15 steps each	training time	§5
PolyPythias	410M, seed 1	init. and data order	§4
OLMo-2-1B	pretraining end, 3 mid-training runs, SFT, DPO, Instruct	continued training	§5
Controlled	4- and 12-layer, C4, 2–3 init.	perturbations, data masking	§3, §6

Table 1: Model populations and the relation between models that each one varies.

trained models. (2) We show that carriers are assigned by a competition among candidate heads during pretraining, and use it to tell when a single-head readout of a data intervention reports a change in the mechanism and when it reports a change of carrier. (3) We provide reliability-corrected comparisons of head maps, a protocol for reusing circuit findings, and an audit that finds four public checkpoints whose lineage differs from their labels (Appendix A).

2 Setup

Relations between models. Two models are *siblings* if they were trained from the same initialization on different data, and *unrelated* if their initializations differ. A model’s *descendants* continue its training: later checkpoints of the same run, mid-training on a new mixture, and SFT, DPO and instruction tuning. A training *lineage* is a model together with its descendants.

Model populations. Table 1 lists the populations and the relation each one varies. DataDecide trains three initializations on 25 corpora at 1B parameters for 100B tokens, and on 10 to 23 corpora at thirteen smaller sizes (Magnusson et al., 2025). We checked every 1B run’s initialization against the released initial weights and use the 69 runs whose initialization matches its label. From Pythia we use the standard and deduplicated models at six sizes from 70M to 12B and fifteen checkpoints each of the 160M and 410M pairs (Biderman et al., 2023), and from OLMo-2-0425-1B the end of pretraining (4T tokens), three midtraining runs on different mixtures (51B tokens each), and the SFT, DPO and Instruct models (Team OLMo et al., 2024). Controlled pretraining uses Llama-style models with 4 or 12 layers trained on C4 (Appendix C).

Head roles. A head role is a score computed for every head, which gives a [layer $\times$ head] role map for each model. We use eight attention roles, namely induction, previous token, attention sink, knowledge retrieval, duplicate token, current token, two tokens back and delimiter (Olsson et al., 2022; Xiao et al., 2024; Wu et al., 2025), and one weight-based role, the copying score of the head’s output-value (OV) circuit (Appendix B).

Causal carriers of IOI. An IOI prompt names an indirect object (IO) and a subject, mentions the subject a second time (S2), and ends at a final position (END) where the model should predict the IO. On 200 prompts (Wang et al., 2023) we compute three quantities for every head. Its direct logit attribution (DLA) is the head’s direct contribution at END to the logit difference between IO and subject, with every normalization layer linearized at its actual scale. Its ablation effect is the drop of the logit difference when the head’s output is replaced by its mean at END. Its attention from END to the IO and S2 tokens completes the profile. The ablation effects of all heads form the *causal map* of a model. A model’s *carriers* are its three heads with the largest DLA, and its *main carrier* is the first of them. Carriers are compared between two models by the overlap of their top three and by *transfer*, the damage that ablating the source model’s carriers does in the target, relative to ablating the target’s own carriers.

Comparing head maps. Two role maps or causal maps M_A and M_B are compared within each layer, so that agreement on the layer that hosts a role does not count as agreement on the head:

$$r_{AB} = \frac{1}{L} \sum_l \rho(M_A^{(l)}, M_B^{(l)}), \quad \tilde{r}_{AB} = \frac{r_{AB}}{\sqrt{R_A R_B}},$$

where $M^{(l)}$ is the row of layer l , ρ is Spearman’s correlation over its heads, and R_A is the split-half reliability of model A ’s map, computed with the same statistic on two halves of the inputs and Spearman–Brown corrected. We report $\tilde{r}_{AB}$ as the correspondence of two maps. The correction matters for causal quantities (Table 2). Most heads contribute nothing to IOI, so a rank correlation of DLA over all heads of a layer orders near-zero values and reaches a reliability of only 0.40–0.50, although the DLA values of the contributing heads replicate at 0.98. Causal maps from ablation are reliable under the rank statistic (0.83–0.86), and

	induction	END→IO	ablation	DLA (rank)	DLA (value)
DataDecide-1B (9 models)	1.00	0.99	0.83	0.50	0.98
Pythia-410M	0.99	0.99	0.83	0.40	0.99
OLMo-2-1B (pretraining end)	1.00	0.99	0.86	0.42	0.98

Table 2: Split-half reliability of head maps (Spearman–Brown corrected). Rank statistics are reliable for attention maps and ablation maps; for direct logit attribution (DLA) the values of the contributing heads are reliable, and the ranks of the many near-zero heads are not.

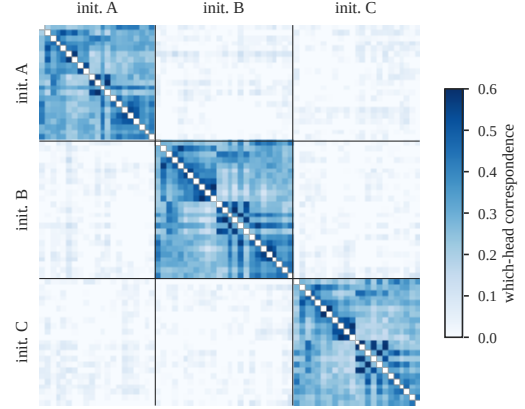


Figure 2: Which-head correspondence between all pairs of the 69 DataDecide-1B models (mean of four attention roles), ordered by initialization and, within each block, by corpus. Models from one initialization form one block across all 25 corpora.

carriers and transfer use the contributing heads directly.

3 Initialization fixes a coordinate system

We call the assignment of roles to head indices within each layer a model’s *coordinate system*. Two models share it to the degree that the same index carries the same role.

Shared across corpora. In the 69 DataDecide-1B models, the profile of every role over layers is as similar between unrelated models as between siblings, and which head of a layer takes the role follows the initialization (Figure 2, Table 3). Ordered by initialization, the pairwise correspondences form three blocks. Siblings correspond at 0.31 on average and unrelated models at 0.00, and the two distributions barely touch (lowest sibling pair 0.10, highest unrelated pair 0.14). Siblings agree on the head for all nine roles (0.18–0.43), and unrelated models agree at chance whether they share the corpus or not. In a per-head variance decomposition, the initialization has a significant effect on 66–79% of heads, the corpus on 4–7%,

Role	which head within the layer			which layer
	same init. diff. corpus	diff. init. same corpus	both differ	same init. / both differ
Induction	0.33	-0.02	0.00	0.49 / 0.41
Previous token	0.42	-0.01	-0.00	0.46 / 0.38
Attention sink	0.18	-0.01	-0.00	0.64 / 0.58
Knowledge retrieval	0.31	-0.00	0.00	0.60 / 0.54
Duplicate token	0.36	-0.02	-0.01	0.62 / 0.57
Current token	0.32	-0.02	-0.00	0.61 / 0.59
Two tokens back	0.43	-0.00	0.01	0.62 / 0.58
Delimiter	0.22	-0.01	0.00	0.57 / 0.50
OV copying (weights only)	0.25	-0.01	-0.01	0.68 / 0.64

Table 3: Correspondence of role maps across the 69 DataDecide-1B models. Which head of a layer takes a role is shared by siblings and by no other pair; the profile over layers is shared by every pair.

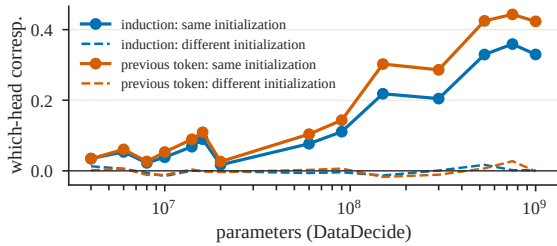


Figure 3: The shared coordinate system across 14 DataDecide sizes: within-layer correspondence of induction and previous-token maps between siblings (solid) and between unrelated models (dashed).

the rate expected by chance. The weight-based OV copying role follows the initialization in the same way, so the shared assignment lies in the weights rather than in the probe inputs.

Across scale and model families. The coordinate system grows stronger with scale (Figure 3). Across the 14 DataDecide sizes, sibling correspondence is small below 20M parameters, clear from 60M, and reaches 0.33–0.44 at 0.5–1B, while unrelated models stay at zero at every size. In Pythia, the standard and deduplicated models correspond at 0.38–0.72 on the induction, previous-token and retrieval maps at each of the six sizes from 70M to 12B, compared with a 95th percentile of 0.04–0.26 for permuted heads.

Set early in training. Controlled pretraining shows when the coordinate system is fixed (Table 4). Noise as large as the weights themselves, added at 1% of training, rearranges which heads take the roles (correspondence with the uninterrupted run 0.36). From 2.5% on, the same perturbation is absorbed and the heads return to their slots (0.85–0.98, compared with 0.97 for an exact rerun).

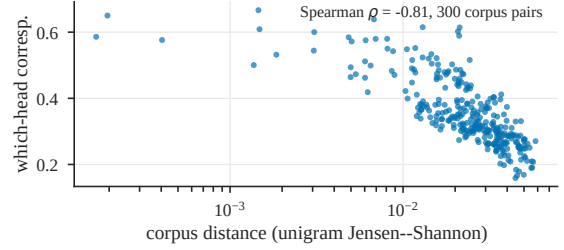


Figure 4: DataDecide-1B siblings share more of the coordinate system when their corpora are closer: which-head correspondence (mean over induction, previous-token and retrieval maps and over the three initializations) against the unigram Jensen–Shannon distance between the two corpora.

Switching the corpus to code after this point keeps a correspondence of 0.60–0.95 with the uninterrupted layout, the more the later the switch. In the public suites, DataDecide-1B siblings already share the coordinate system at 3.6% of training as strongly as at the end.

Shared in proportion to corpus similarity. How much of the coordinate system two siblings share depends on their data (Figure 4). Over the 300 pairs of DataDecide corpora, siblings share more of the layout the closer their corpora are in word distribution (Spearman $\rho = -0.81$), from 0.16 for the most distant corpora to 0.67 for the closest.

In two siblings, then, “layer 12, head 3” refers to heads with the same roles, and their components can be compared head by head without alignment, across corpora that range from web crawls to curated and quality-filtered mixtures.

4 Training assigns the causal carriers

Siblings share roles, not carriers. The causal carriers of a task are not part of the shared coordinate system (Figure 1). For IOI, the attention from END to the IO token corresponds between DataDecide siblings at 0.24, at the level of the roles in §3. Their causal maps correspond at 0.02 after reliability correction, as low as between unrelated models (0.01), and the carriers of one sibling transfer 3% of the damage to another. None of the 18 sibling pairs among four DataDecide corpora shares its main carrier. Comparing DLA values instead of ranks gives the same result, with siblings at 0.11 and unrelated models at 0.07–0.10. Over all 69 DataDecide-1B models, the uncorrected correspondences show the same contrast: the END→IO attention maps of

Step	Share	300-step warm-up				1,000-step warm-up		Optimizer reset
		Switch to code	Noise 100%	Noise 10%	Code, 12 layers	Switch to code	Noise 100%	Noise 100%
0	0%	-0.03	0.03	0.74	-0.03	-	-	-
100	1%	0.09	0.36	0.94	0.40	-	0.16	-
250	2.5%	0.60	0.85	0.97	-	0.41	0.70	0.81
500	5%	0.72	0.96	0.99	-	-	0.87	0.91
1000	10%	0.71	0.95	0.99	0.84	0.69	0.87	0.97
2000	20%	0.81	0.98	1.00	-	-	0.94	-
4000	40%	0.86	0.96	1.00	-	-	-	-
8000	80%	0.95	0.98	1.00	-	-	-	-
exact rerun					0.97			

Table 4: When the coordinate system is set (controlled 4-layer models, means over two initializations; one 12-layer model for the code switch in the fourth column; Appendix C). Which-head correspondence (mean of induction and previous-token maps) between the final layout of a run that was intervened on at a given step and the uninterrupted run: switching the corpus to code, or adding Gaussian noise scaled to 100% or 10% of each tensor’s standard deviation, under the default 300-step warm-up, a 1,000-step warm-up, and with a freshly initialized optimizer. Step 0 for the code switch: the same initialization trained on code from scratch.

siblings correspond at 0.29 and those of unrelated models at 0.00, while the causal maps correspond at 0.07 between siblings and at 0.00–0.01 otherwise, and a sibling’s three to twenty strongest heads transfer 7–16% of the damage, compared with 4–12% between unrelated models.

Independent of data similarity. Carriers differ between siblings however similar their data are. Carrier transfer between DataDecide siblings barely depends on the distance of their corpora (Spearman $\rho = -0.10$), whereas the shared coordinate system follows it closely ($\rho = -0.81$). Removing the Flan instruction data, about 1.5% of the Dolma corpus, gives the three sibling pairs with the most similar role maps of all the pairs we compare (0.57–0.58), and each of them carries IOI on different main carriers. The standard and deduplicated Pythia models carry IOI on largely the same heads at 160M and 1B (transfer 0.85 and 0.90) and on different heads at 410M, 1.4B, 6.9B and 12B (−0.05 to 0.16; Table 5), and at 410M their attention maps correspond at 0.48 while their causal maps correspond at 0.15.

Same algorithm, different heads. The siblings differ in which heads carry IOI, and agree on how it is computed. In the five Pythia models we analysed head by head (160M and 410M, standard and deduplicated, and a 410M model from another initialization), IOI runs on the same functional classes (Wang et al., 2023): a subject-suppression head that attends to S2 and writes against it through a negative OV copying circuit, and several name movers

Size	main carrier (standard)	also in sibling’s top three	top-3 shared	transfer
160M	8.9	yes	2/3	0.85
410M	11.4	no	0/3	0.05
1B	12.1	no	0/3	0.90
1.4B	10.7	no	0/3	-0.05
6.9B	20.28	no	0/3	0.04
12B	20.11	no	0/3	0.16

Table 5: IOI carriers of standard and deduplicated Pythia models, which share their initialization. Transfer: damage of the standard model’s carriers in the deduplicated model, relative to the latter’s own carriers.

that attend to the IO and copy it (Table 6). Each class has several members in each model, and the 410M siblings give the main role to different members (Figure 5). The standard model’s main carrier, 11.4, contributes 1.16 to the logit difference and 0.02 in the deduplicated model, whose own main carrier, 12.12, contributes 0.97 there and 0.00 in the standard model. Each is nearly silent in the other model and attends to the first token instead.

Carriers are found causally. Because several heads of a class share its attention profile, the profile does not single out the member that carries the task (Table 7). Mapping the standard model’s carriers onto a target by head index, by the nearest attention and OV profile, or by a rule based on the class signature recovers the target’s carriers for some pairs and misses them for others. A mapping that includes a causal readout in the target recovers them for every pair, at 85–94% of the damage of the target’s own carriers.

Model	subject-suppression head				name movers	negative name movers
	head	effect	attn. to S2	OV copy		
Pythia-160M	8.9	1.43	0.94	-1.52	4	1
Pythia-160M-dedup.	8.9	1.65	0.90	-1.36	3	2
Pythia-410M	11.4	1.11	0.39	-1.81	14	2
Pythia-410M-dedup.	12.12	0.95	0.34	-1.83	11	3
Pythia-410M seed 1	12.2	2.28	0.54	-1.99	12	5

Table 6: The IOI algorithm in five Pythia models. Every model has a subject-suppression head, which attends to the repeated subject (S2) and copies it negatively (OV copying score below zero), together with several name movers and negative name movers; which head suppresses the subject differs between the 410M siblings. Effect: direct effect on the logit difference on held-out prompts.

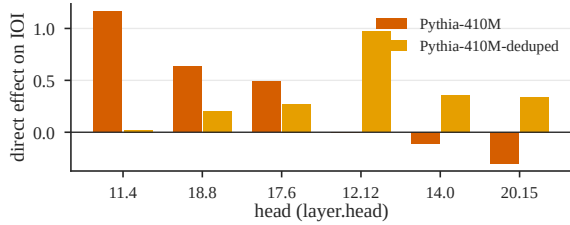


Figure 5: Same algorithm, different carriers. Direct effect on the IOI logit difference of the carriers of Pythia-410M and of its deduplicated sibling, measured in both models. Each model’s main carrier is silent in the other.

5 Carriers move during pretraining, then persist

During pretraining. Within one run, the carriers change during most of pretraining (Figure 6). In Pythia-410M, IOI appears at step 3000. Head 11.4 becomes the main carrier at step 2000 and keeps the role, while the second and third places change hands until step 66k, after which the final carriers are in place. The deduplicated sibling settles on a different main carrier, 12.12, by step 6k and completes its final carriers at step 100k. From the moment carriers appear, the two 410M siblings share none of them, while their attention roles start at a correspondence of 0.79 and diverge slowly to 0.47. The 160M siblings instead take the same main carrier, 8.9, at step 3000, keep it to the end, and transfer 69–99% of their damage to each other from then on, matching their shared carriers in Table 5. In DataDecide-1B, the main carrier at the point where IOI first appears is still the main carrier at the end of training in three of twelve runs, and in the other nine another head takes over later in pretraining.

Mapping	same initialization		different initialization
	160M-dedup.	410M-dedup.	410M seed 1
Same head index	0.87	0.06	0.00
Nearest attention / OV profile	0.92	0.84	-0.06
Class rule (attention and OV sign)	0.92	0.18	1.06
Profile with the target’s causal readout	0.89	0.85	0.94

Table 7: Mapping the standard Pythia model’s carriers onto a target model: damage of the mapped heads in the target relative to the target’s own carriers, on held-out prompts. A mapping that uses a causal readout in the target recovers the carriers for every pair.

	attention role	ablation map	top-3 carriers	transfer	main carrier
Pretraining end → midtraining, 5B tokens	0.96	0.85	2/3	0.93	14.4
Pretraining end → midtraining mix 1, 51B	0.96	0.84	2/3	0.94	14.4
Pretraining end → midtraining mix 2, 51B	0.96	0.85	2/3	0.95	14.4
Pretraining end → midtraining mix 3, 51B	0.96	0.88	2/3	0.94	14.4
SFT → DPO	0.99	1.00	3/3	1.00	14.4
DPO → Instruct	1.00	1.00	3/3	1.00	14.4
Pythia-410M step 66k → 143k	0.92	0.78	3/3	0.99	11.4
Pythia-410M vs. its deduplicated sibling	0.47	0.15	0/3	0.09	12.12

Table 8: IOI carriers along a training lineage (OLMo-2-1B, Pythia-410M) and between the 410M siblings. Attention: END→IO map; ablation: reliability-corrected causal map; transfer: damage of the earlier model’s carriers in the later model, relative to its own; main carrier: the later model’s.

After pretraining. Descendants keep the carriers that pretraining settled (Table 8). From the end of OLMo 2 pretraining to the end of each of three midtraining runs on different mixtures, the causal maps correspond at 0.84–0.88, two of the three carriers are retained, and the earlier carriers produce 94–95% of the later model’s own damage. The main carrier, 14.4, is the same in all of them. SFT, DPO and instruction tuning keep the same three carriers with full transfer, and so does the second half of Pythia-410M pretraining. A finding about the heads that carry IOI in a base model is also a finding about its midtrained, preference-tuned and instruction-tuned descendants, in line with the observation that fine-tuning enhances existing mechanisms (Prakash et al., 2024).

Carriers are assigned in each run’s own pretraining: siblings go through separate competitions among candidate heads, and descendants inherit the outcome, which gives the two halves of Figure 1.

6 Data interventions and the competition among candidates

The competition among candidates also decides what a training-data intervention does to a single head. Training-data attribution identifies the

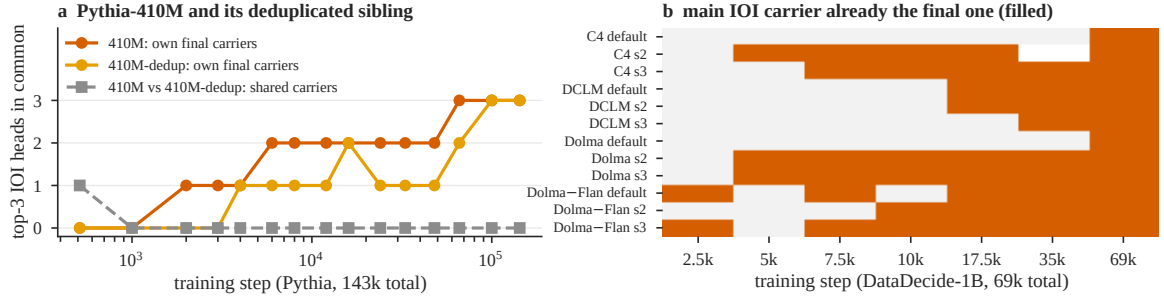


Figure 6: Carriers during pretraining. (a) Number of each Pythia-410M model’s final carriers already among its three strongest heads (solid), and number of carriers the two siblings share (dashed). (b) Twelve DataDecide-1B runs (four corpora $\times$ three initializations), filled where the current main carrier is the run’s final main carrier.

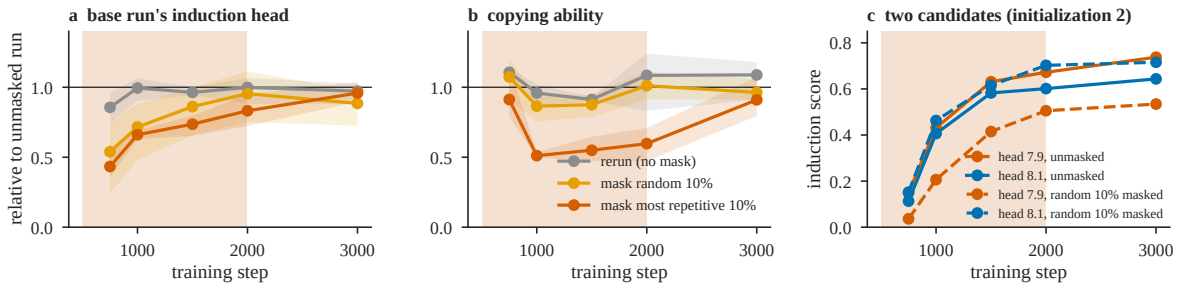


Figure 7: Masking training data while induction heads emerge (shaded: steps 500–2000) in controlled 12-layer models from three initializations. (a) Induction score of the unmasked run’s strongest induction head, relative to the unmasked run (mean and range over initializations); (b) copying ability (loss reduction on the repeated half of random sequences), relative to the unmasked run. (c) Initialization 2 has two near-equal candidates, and masking a random 10% of sequences gives the role to the second one from the moment it emerges.

data that create a component by removing data, retraining, and examining the same head (Chen et al., 2026). We run such interventions in controlled pretraining, where the initialization and the data order are fixed and only the intervention differs (Figure 7). From three initializations we mask the loss of 10% of the sequences during steps 500–2000, the window in which induction heads emerge, choosing either the most repetitive sequences, which most accelerate induction heads (Chen et al., 2026), or a random tenth.

One leading head. When one head leads the role, the intervention delays that head. Masking the most repetitive data lowers the strongest head’s induction score to 36–55% of the unmasked run at step 750 and copying ability to about half at step 1000. Both recover after the window, and at step 3000 the strongest induction head is the same head in all three initializations, as it is in a rerun of the unmasked configuration. A drop in the score of that head is a delay of the mechanism, which is what a single-head readout reports.

Two close candidates. When two heads compete, a small change in the data decides the winner. In initialization 2, heads 7.9 and 8.1 rise together, and 7.9 stays slightly ahead in the unmasked run and in its rerun. Masking a random 10% of the sequences lets 8.1 lead from the moment induction appears, and it keeps the role. At step 3000 the original head scores 72% of the unmasked run, while the strongest induction score (97%) and copying ability (92%) are intact. A readout of 7.9 alone would report a weakened mechanism where the mechanism has moved to another head. In the two initializations with a clear leader, the same random mask leaves the leading head in place (95% and 98% of its unmasked score) and copying ability intact (91% and 106%). Induction heads show the pattern of the IOI carriers in a fully controlled setting: a role with one leading candidate stays on its head, and a role with several candidates goes to whichever candidate the data of the run favour while the role forms.

7 Related Work

Universality and stability of components.

Whether circuits recur across models has been studied across training and scale, where the algorithm persists while its heads change (Tigges et al., 2024), and across seeds, where attention heads, most of all those in middle layers, are unstable, and decoupled weight decay makes them more stable (Bali et al., 2026), and where heads with a shared role are positionally inconsistent (Trott, 2025). Representations of independently trained networks share structure up to rotation or permutation (Li et al., 2016; Kornblith et al., 2019; Ainsworth et al., 2023). Most seed variants, among them the PolyPythias (van der Wal et al., 2025) and MultiBERTs (Sellam et al., 2022), change initialization and data order together. Separating the initialization from the data, and attention roles from causal carriers, shows why components agree between siblings in one sense and not in the other.

Circuit discovery. Circuits are discovered by path patching and its automated variants (Wang et al., 2023; Conmy et al., 2023), and universality has been examined in small algorithmic models and at the level of neurons (Chughtai et al., 2023; Gurnee et al., 2024). Our results identify which statements of a discovered circuit can be reused in a related model, and show that the rest require a causal readout in that model.

Mechanisms after further training. Fine-tuning reuses and enhances the circuits of the base model (Prakash et al., 2024). We trace carriers through a complete public lineage, including mid-training on new mixtures, SFT, DPO and instruction tuning, and contrast it with siblings trained separately. Tigges et al. (2024) observe that the heads of a circuit change over the course of training while its algorithm persists. Our lineage results place these changes in pretraining and show that they stop once pretraining has settled the carriers.

Training data and components. Influence-based attribution traces heads to training samples and verifies the attribution by retraining (Chen et al., 2026), and properties of the data such as burstiness determine when in-context learning emerges (Chan et al., 2022). Our controlled masking distinguishes interventions that delay a head from those that move its role to another candidate.

Reliability of circuit measurements. Circuit scores vary across inputs and methods (Méloux et al., 2026), and redundant heads repair ablations (McGrath et al., 2023). Split-half reliability of every head map, and the correction of cross-model correspondence by it, make causal maps comparable across models.

8 Discussion

A protocol for reusing a circuit. To apply a circuit found in model A to model B: (i) establish the lineage from the weights, since released labels can be wrong (Appendix A); (ii) if B descends from A, reuse the named heads and confirm them with a single ablation of the named set; (iii) if B is a sibling of A, compare role maps head by head to inherit role labels, and localize the task’s carriers with single-head ablations in B; (iv) otherwise, transfer the algorithm and the layer profile and run circuit discovery in B; (v) in every case, report the split-half reliability of the head maps being compared.

Data attribution and model diffing. A study that retrains a model under a data intervention and tracks a named head compares two siblings with the same initialization and data order. Our controlled runs show that such a comparison reads the mechanism correctly when one head leads the role, and that a role with close candidates can move under an intervention as small as a random 10% mask. Checking the strongest head of the role and the behaviour alongside the named head separates a delayed mechanism from a moved one. Comparisons between a base model and its fine-tuned or instruction-tuned descendants, as in model diffing, fall on the other side of the rule: the descendants keep the base model’s carriers, so differences between them can be attributed to the same named heads.

Coordinates and carriers. The initialization determines where each role can be carried, and the data of a run determine which candidate carries it. The two are settled at different times: the coordinates within the first few percent of training (Table 4), before the corpus has had much effect, and the carriers late in pretraining (Figure 6), after each corpus has shaped the competition. This is why the coordinate system tracks the similarity of the corpora and the carriers do not. The variation that Tigges et al. (2024) and Bali et al. (2026) observe across seeds arises in this competition.

Limitations

We trace one task circuit, IOI, through its causal carriers, and nine roles through attention and weights; extending the carrier analysis to circuits for factual recall and in-context learning is the natural next step. The public lineages we use reach 12B parameters in Pythia and 1B in OLMo 2, and the controlled interventions use models with up to 12 layers, so the same measurements on larger lineages with released midtraining and post-training stages would carry the rule to current frontier pipelines.

Ethical Considerations

This work analyses publicly released models and data. It reports lineage mismatches in public checkpoints so that users of these suites can compare the models they intend to compare.

References

Samuel K. Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. 2023. [Git re-basin: Merging models modulo permutation symmetries](#). In *International Conference on Learning Representations*.

Karan Bali, Jack Stanley, Praneet Suresh, and Danilo Bzdok. 2026. [Quantifying LLM attention-head stability: Implications for circuit universality](#). In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 5911–5944.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. [Pythia: A suite for analyzing large language models across training and scaling](#). In *Proceedings of the 40th International Conference on Machine Learning*.

Stephanie C. Y. Chan, Adam Santoro, Andrew K. Lampinen, Jane X. Wang, Aaditya Singh, Pierre H. Richemond, Jay McClelland, and Felix Hill. 2022. [Data distributional properties drive emergent in-context learning in transformers](#). In *Advances in Neural Information Processing Systems*.

Jianhui Chen, Yuzhang Luo, and Liangming Pan. 2026. [Mechanistic data attribution: Tracing the training origins of interpretable LLM units](#). In *Proceedings of the 43rd International Conference on Machine Learning*.

Bilal Chughtai, Lawrence Chan, and Neel Nanda. 2023. [A toy model of universality: Reverse engineering how networks learn group operations](#). In *Proceedings of the 40th International Conference on Machine Learning*.

Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. 2023. [Towards automated circuit discovery for mechanistic interpretability](#). In *Advances in Neural Information Processing Systems*.

Wes Gurnee, Theo Horsley, Zifan Carl Guo, Tara Rezaei Kheirkhah, Qinyi Sun, Will Hathaway, Neel Nanda, and Dimitris Bertsimas. 2024. [Universal neurons in GPT2 language models](#). *Transactions on Machine Learning Research*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. [Similarity of neural network representations revisited](#). In *Proceedings of the 36th International Conference on Machine Learning*.

Yixuan Li, Jason Yosinski, Jeff Clune, Hod Lipson, and John Hopcroft. 2016. [Convergent learning: Do different neural networks learn the same representations?](#) In *International Conference on Learning Representations*.

Ian Magnusson, Nguyen Tai, Ben Bogin, David Heine-man, Jena D. Hwang, Luca Soldaini, Akshita Bhagia, Jiacheng Liu, Dirk Groeneveld, Oyvind Tafjord, Noah A. Smith, Pang Wei Koh, and Jesse Dodge. 2025. [DataDecide: How to predict best pretraining data with small experiments](#). In *Proceedings of the 42nd International Conference on Machine Learning*.

Thomas McGrath, Matthew Rahtz, János Kramár, Vladimir Mikulik, and Shane Legg. 2023. [The hydra effect: Emergent self-repair in language model computations](#). arXiv preprint arXiv:2307.15771.

Maxime Méroux, François Portet, and Maxime Peyrard. 2026. [Mechanistic interpretability as statistical estimation: A variance analysis](#). In *Proceedings of the 43rd International Conference on Machine Learning*.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, and 7 others. 2022. [In-context learning and induction heads](#). *Transformer Circuits Thread*.

Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. 2024. [Fine-tuning enhances existing mechanisms: A case study on entity tracking](#). In *The Twelfth International Conference on Learning Representations*.

Thibault Sellam, Steve Yadlowsky, Jason Wei, Naomi Saphra, Alexander D’Amour, Tal Linzen, Jasmijn Bastings, Iulia Turc, Jacob Eisenstein, Dipanjan Das, Ian Tenney, and Ellie Pavlick. 2022. [The MultiBERTs: BERT reproductions for robustness analysis](#). In *International Conference on Learning Representations*.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groen-
evelt, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling
Gu, Shengyi Huang, Matt Jordan, Nathan Lambert,
Dustin Schwenk, Oyvind Tafjord, Taira Anderson,
David Atkinson, Faeze Brahman, Christopher Clark,
Pradeep Dasigi, Nouha Dziri, and 24 others. 2024. [OLMo 2 furious](#). arXiv preprint arXiv:2501.00656.

Curt Tigges, Michael Hanna, Qinan Yu, and Stella Bi-
derman. 2024. [LLM circuit analyses are consistent
across training and scale](#). In *Advances in Neural
Information Processing Systems*.

Sean Trott. 2025. [Toward a theory of generalizability
in LLM mechanistic interpretability research](#). arXiv
preprint arXiv:2509.22831.

Oskar van der Wal, Pietro Lesci, Max Muller-Eberstein,
Naomi Saphra, Hailey Schoelkopf, Willem Zuidema,
and Stella Biderman. 2025. [PolyPythias: Stability
and outliers across fifty language model pre-training
runs](#). In *International Conference on Learning Rep-
resentations*.

Kevin Ro Wang, Alexandre Variengien, Arthur Conmy,
Buck Shlegeris, and Jacob Steinhardt. 2023. [Inter-
pretability in the wild: a circuit for indirect object
identification in GPT-2 small](#). In *International Con-
ference on Learning Representations*.

Wenhao Wu, Yizhong Wang, Guangxuan Xiao, Hao
Peng, and Yao Fu. 2025. [Retrieval head mechanis-
tically explains long-context factuality](#). In *Interna-
tional Conference on Learning Representations*.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song
Han, and Mike Lewis. 2024. [Efficient streaming lan-
guage models with attention sinks](#). In *International
Conference on Learning Representations*.

A Lineage of public checkpoints

Head layouts reveal which checkpoints share an ini-
tialization, and we confirmed every case with the
weights. In DataDecide-1B, six runs were trained
from initializations other than the one their seed
label names. They form three pairs, each sharing
an unregistered initialization: their initial weights
correlate at 0.000 with those of the labelled seed,
and their weights at step 2500 correlate within each
pair as strongly as those of true siblings. We ex-
clude these six runs from all comparisons by ini-
tialization. The PolyPythias 160M weight-seed
variants were trained from the standard initializa-
tion: their weights at step 1 correlate at 1.0 with
the standard model’s initial weights, although their
released initial weights differ. The released final
checkpoint of Pythia-2.8B-deduped is nearly iden-
tical to Pythia-2.8B (weight correlation 0.99), so
the two cannot serve as siblings and Pythia compar-
isons use the other six sizes. The main revision of

OLMo-2-0425-1B comes from a different training
run than the released stage-1 and stage-2 check-
points and the SFT, DPO and Instruct models. A
layer-5 query matrix of main correlates at 0.000
with every stage checkpoint and with the Instruct
model, while the Instruct model correlates at 0.976
with the end of midtraining, and the head layouts
agree. Lineage analyses of OLMo-2-1B start from
the stage-2 checkpoints.

B Head roles

Induction is the attention from the second copy
of a repeated random block to the token after the
earlier occurrence. Previous token, current token
and two tokens back are the attention to positions
 $t - 1$, t and $t - 2$ on natural text, and the attention
sink is the attention to position 0. Knowledge re-
trieval is the attention from the final prompt token
to the answer entity in knowledge-conflict prompts.
Duplicate token is the attention from the second
copy of a random block to the same token’s first
occurrence, and delimiter is the attention mass on
punctuation and newline tokens. OV copying is
 $\sum_i \text{Re } \lambda_i / \sum_i |\lambda_i|$ over the eigenvalues λ_i of the
full OV circuit from the embedding to the unem-
bedding. Every attention role is averaged over two
halves of its inputs, with split-half reliabilities of
0.98 or higher.

C Controlled pretraining

All controlled models are Llama-style decoders
with a 50k vocabulary trained on C4 with sequences
of 512 tokens and batches of 64 sequences, using
AdamW with learning rate 10^{-3} , 300 warm-up
steps and cosine decay over 10k steps. The inter-
ventions of Table 4 use 4-layer models with 8 heads
and width 256, trained for 10k steps from two ini-
tializations. A run is branched from its saved state
at a given step, and either Gaussian noise scaled
to each tensor’s standard deviation is added to the
weights or the corpus is switched to code; the final
layout is compared with that of the uninterrupted
run. One condition repeats the code switch with
a 12-layer model, and two conditions lengthen the
warm-up to 1,000 steps or start the branch with a
fresh optimizer state. The masking experiments of
§6 use 12-layer models with 12 heads and width
768, trained for the first 3000 steps of the same
schedule from three initializations with one data
order. Masking sets the loss weight of the selected
sequences to zero for steps 500–2000 and leaves

the data order unchanged. A sequence's repetition score is the share of tokens whose two preceding tokens occurred earlier in the sequence followed by the same token, and the masking threshold is its 90th percentile on C4 (0.08). Random masking selects a tenth of the sequences with a separate random generator.